Thesis: defined bull put spreads(long put further OTM and sell put closer to underlying)

Theorized edge comes from this : reversion- some categories of news cause selling that is larger than the actual fundamental damage, so price is technically not representative of the true value of company

-Volatility premium : IV spikes after a shock and sometimes can lag actual volume. The premium is more than expected but this also means the OTM put can be more expensive as well.

Trading Rules(this should change overtime as we perform backtests and adjust different factors)

1. Bull put spreads only to avoid uncapped tail risk
2. Within DTE we should try to avoid earnings events
3. Liquid names only -> decent bid ask spreads and good open interest
4. Strike placement: try to find out what is optimal for this →initial idea is to have short strike below post announcement low at a set delta range(maybe 10-30)
5. Expiry window → window should avoid earnings but maybe 2 - 5 weeks for most trades
6. Max loss per trade→ 1-3% of allocated capital
7. Wait for stock to settle before entering as IV should lag actual volume

Team Roles:

We need roles to better help what each of us are gonna work on. This is what I can think of that we need right now

Signal Research →(news LLM) so categorizing events, determining if change fundamentals, testing LLM calls to correctly categorize items
-some ideas : anonymize events to avoid backtest bias and use tests to see if LLM truly doesn't recognize company

-we also could find what type of news events historically lead to the most reversions

Data Backtest Engineering →
-find sources of data using what we are given and free stuff online
- price/options/ IV data pipelines, time correctness

- simulating fills and cleaning data for signal research as well
- Finding what rules and logic return the best

Actual logic risk/execution →

Sizing, risk limits, exits, programming how we are filling orders in a bid ask scenario, filling the orders autonomously, screening, making sure there's no bugs in the execution itself(double fills, not recognizing previous orders, not recognizing positions are open/close after placing limit orders etc.)

What im thinking for research rn

- First we need to build a set of non earnings shocks and tag each with category and measure abnormal returns at different time periods(post 1 day,5 day, 10 day ,etc.)
- Track where IV is lagging behind realized vol(where selling cools but IV doesn't proportionally fall with)
- Based on this, we can simulate put spread and see what rules work best and what technical indicators to combine or which categories return the best

Data we need :

Intraday stock prices/volume, timestamped historical news and associated reactions, earnings calendar(historical and present), historical options chain, historical IV, and for live testing we need current options quotes and paper trading.

Some categories:
Analyst downgrade, peer/sector negative news, lawsuit filed, regulatory probe, short seller report, product recall, dilution/secondary offering

Some news events should fall out of these categories and there would be special cases where each would definitely impact the company and others not. There also would be cases where it would impact the company but not as much as the stock reaction reflects→

Parameters we can test

Entry lag(x amount of days or settled trigger from volume)
Short strike delta(ex: 10-30)
Spread width(how big the difference between our strikes are)
DTE at entry->14,21,more etc.
Profit target→Immediate exits at certain thresholds or exit if we return large portion in small amount of time

Stop loss→ none , 1.5-3x credit or dynamic
Filters→ market cap, interest, volume, IV rank, technical indicators etc.